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TEAM NAME : **TEAM TARGARYENS**

COLLEGE NAME : **S**ri **V**enkateswara **C**ollege of **E**ngineering

PROBLEM STATEMENT : Many people face delays in receiving proper healthcare due to difficulty identifying the right doctor and slow emergency response. An integrated AI-based platform can help provide faster diagnosis and medical assistance

PROBLEM STATEMENT

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01
Healthcare Accessibility Issues

In many regions of India, people face difficulty accessing timely and reliable healthcare services. Patients often cannot quickly determine the correct medical specialist based on their symptoms, which causes delays in diagnosis and treatment.

02
Emergency Response Delays

During medical emergencies, delays in contacting ambulances, sharing location details, and receiving immediate medical assistance can significantly increase health risks and reduce the chances of survival.

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03
Lack of Integrated Healthcare Platforms

Most existing healthcare services operate separately for doctor consultation, ambulance services, and medicine access. There is no unified system that connects these services into a single efficient healthcare platform.

04
Limited Early Health Assessment Tools

Many patients are unable to identify potential health problems early because they lack access to tools that can analyze symptoms or visible health issues. This often leads to delayed medical consultation and worsening of the condition.

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SOLUTION

AI-Based Disease Detection and Smart Doctor Recommendation

- Users enter symptoms or upload images of health issues through the application.
- Images are processed using OpenCV and analyzed with Convolutional Neural Networks in TensorFlow.
- The AI model predicts possible diseases based on symptoms and image data.
- The system then recommends the most suitable doctor specialization.

Emergency SOS with Family Alerts and Ambulance Tracking

- The application includes an SOS button for emergency situations.
- When activated, it captures the user's live location using GPS.
- The location is sent to nearby ambulance services and registered family members.
- Users can track ambulance movement in real time using backend services like FastAPI.

AI First Aid Guidance and Telemedicine Consultation

- The system provides AI-based first aid guidance based on the user's symptoms or detected health condition.
- It generates step-by-step instructions to help users handle emergencies before medical help arrives.
- This support can help stabilize the patient and reduce risk during critical situations.
- Users can also consult doctors remotely through video calls or chat for immediate medical advice.

Integrated Healthcare Platform with Medicine Ordering

- The platform integrates disease detection, consultation, and emergency services in one system.
- Users can book doctor appointments directly through the application.
- Prescribed medicines can be ordered from nearby pharmacies.
- All user data and orders are securely stored in Firebase.

FLOW DIAGRAM

1.USER LAYER (MOBILE APPLICATION - FLUTTER) :

- Enter health symptoms (fever, cough, chest pain)
- Upload images (skin infection, wound, swelling)
- Press SOS Emergency button
- Request Doctor consultation
- Order medicines

2.BACKEND PROCESSING LAYER (FASTAPI) IS :

- Receive symptom data & images
- Handle SOS requests
- Manage doctor recommendations
- Send notifications to Ambulance & Family
- Connect with Database
- Send responses to Mobile App

3.AI ANALYSIS LAYER(INTERPROCESSING LAYER) :

Symptom analysis → ML Model predict disease & suggest doctor's specialization

Symptom analysis → Image disease detection → OpenCV: Image preprocessing(resize,noise, removal,normalization)

CNN: Analyze processed image for disease detection(TenserFlow)

3.OUTPUT LAYER:

- Predicted disease results
- Recommended doctor
- First aid suggestions
- Ambulance tracking information
- Medicine order confirmation

4.DATABASE LAYER :

- User profile
- Symptom history
- Medical images
- Emergency alerts
- Medicine orders
- Doctor records

5. BACKEND RESPONSIBILITIES :

- User Recommendations
- Symptom History
- Medical Images upload by users
- Medicine Ordering from Nearby Pharmacies

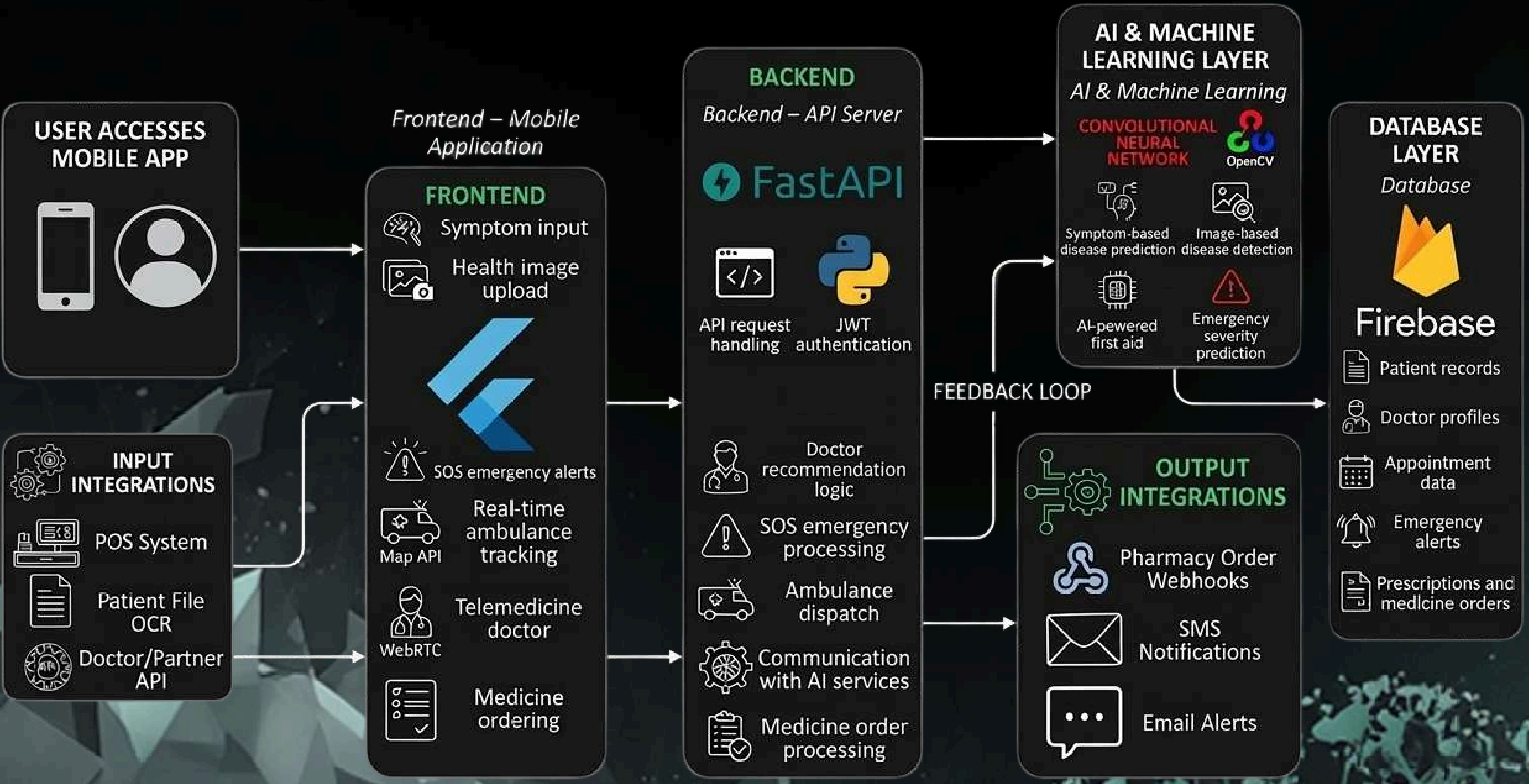
6. HEALTHCARE SERVICE LAYER:

- Smart Doctor Recommendation
- Telemedicine Consultation
- AI-based First Aid Guidance
- Medicine Ordering from Nearby Pharmacies

7.Emergency response system:

SOS Pressed → user location captured → send location to ambulance and family members → notify registered family members → ambulance assigned → Real-Time Ambulance Tracking

TECHNICAL REQUIREMENTS



NOVELTY / USE CASE

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01

Multimodal AI Disease Detection

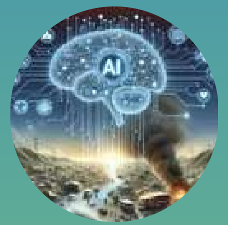
The system analyzes both user symptoms and uploaded images of visible health issues to predict possible diseases, improving accuracy compared to traditional symptom checkers.

02

AI-Based Doctor Specialization Recommendation

Based on the predicted condition, the application automatically recommends the most appropriate medical specialist, helping patients quickly identify the correct doctor.

03



AI-Assisted Emergency Detection

The system can identify potential emergency conditions from user symptoms and suggest immediate activation of emergency services.

04

Smart SOS Emergency System with Real-Time Location Sharing

When the SOS feature is activated, the app sends an emergency alert. It shares the user’s live location with nearby ambulance services and family members for faster response. If internet is unavailable, the system uses the cellular network (SMS/GSM) to send GPS coordinates. This ensures reliable location sharing even in low-network or offline situations

05

AI-Powered First Aid Guidance

The application provides step-by-step first aid instructions based on the user’s symptoms or emergency condition, helping users take immediate action before medical help arrives.

06

Integrated Healthcare Service Platform

Unlike existing systems that offer separate services, this platform integrates disease detection, doctor consultation, ambulance services, medicine ordering, and emergency assistance into a single unified application.

07

Early Health Risk Identification

The AI system helps users identify potential health issues at an early stage, encouraging timely medical consultation and preventing serious complications.

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BUSINESS MODEL / TARGET MARKET

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Revenue can be generated through online doctor consultations conducted via the platform.

Additional revenue can be generated by offering subscription plans that provide advanced health monitoring and priority medical consultations.

The platform can earn service charges when users request emergency ambulance bookings.

Healthcare providers can partner with the platform to promote their services and reach a larger patient base, generating partnership revenue.

User Health Data Input → AI Disease Prediction → Doctor Consultation System → Emergency Assistance → Medicine & Pharmacy Services

- Consultation Fees
- Ambulance Booking Charges
- Pharmacy Partnerships
- Premium Healthcare Services

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